Ultrasound as a predictor of time-loss injury for the patellar tendon, Achilles tendon and plantar fascia in division I collegiate athletes

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ABSTRACT

Objectives Tendinopathy and fasciopathy are common conditions that can result in time-loss injury in athletes. This study aimed to determine if preseason sonographic abnormalities of the patellar tendon, Achilles tendon and plantar fascia are associated with future time-loss injuries in collegiate athletes.

Methods National Collegiate Athletic Association Division I athletes from three institutions participated in this 3-year prospective, observational study. Each athlete completed a questionnaire, self-reporting current and prior symptoms and underwent an ultrasound examination of bilateral patellar tendons, Achilles tendons and plantar fasciae during annual preseason examinations. Ultrasound evaluations assessed for tendon and fascia thickening, hypoechogenicity and neovascularisation. Athletes were monitored for time-loss injury over the subsequent year.

Results A total of 695 athletes across 18 sporting disciplines (61.1% female, age 20.0±1.6, body mass index 23.1±2.9 kg/m^2) were analysed over 3 years. Sonographic abnormalities were identified in 36.6%, 7.5% and 2.8% of the patellar tendons, Achilles tendons and plantar fasciae, respectively. Injuries were reported in 3.3%, 1.6% and 0.7% of these structures with an adjusted relative risk of injury increased by 8.9 (95% CI 3.7, 21.4), 18.8 (95% CI 7.2, 48.8) and 21.0 (95% CI 6.4, 68.1) times in those with preseason ultrasound abnormalities (p<0.001). The presence of an ultrasound abnormality was more predictive of future injury than self-report of a prior injury or pain in the area at the time of the scan.

Conclusion Preseason sonographic abnormalities of the patellar tendon, Achilles tendon or plantar fascia are associated with a higher risk of developing time-loss injuries in collegiate athletes.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Achilles and patellar tendons with sonographic abnormalities are more likely to progress to pain and injury than structurally normal tendons. A history of prior injury is also associated with future injury risk for these structures.

WHAT THIS STUDY ADDS

⇒ This prospective study of collegiate athletes demonstrates that sonographic abnormalities of the Achilles tendon, patellar tendon or plantar fascia are independently associated with an increased risk of time-loss injury. The risk associated with sonographic abnormalities appears to be higher than that with self-reported pain or prior injury at the time of scanning. Though the negative predictive value of a normal ultrasound examination is high, the positive predictive value of an abnormal ultrasound examination is low, suggesting that not all athletes with sonographic abnormalities will actually become injured.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ A preseason ultrasound evaluation of the Achilles tendon, patellar tendon and plantar fascia may identify at-risk athletes, particularly in the collegiate athlete population. A normal ultrasound is highly correlated to staying injury-free for that particular structure during the following year.

INTRODUCTION

Patellar tendinopathy, Achilles tendinopathy and plantar fasciopathy are common conditions that have the potential to affect athletic performance and may result in time-loss injury limiting athlete participation in training, practice and competition.[1–3] Injuries related to tendinopathy/fasciopathy occur due to repeated overuse, increased tendon loading and impaired recovery processes.[4] Numerous studies have evaluated risk factors related to lower extremity tendon/fascia injuries in athletes, with prior injury to the area being one of the most common.[5–7] Additional risk factors include training volume, altered biomechanics and demographic characteristics such as age and body mass index.[4 8–10]

Tendinopathy and fasciopathy are diagnosed based on history and physical examination and can incorporate imaging. Ultrasound is a cost-effective, accessible diagnostic tool with real-time interpretation commonly used in the clinical evaluation of these three conditions[11–13] and displays similar levels of accuracy to MRI.[14 15] It is clear that ultrasound abnormalities may be present in asymptomatic individuals, but systematic review-level evidence demonstrates that those with asymptomatic abnormalities are much more likely to progress to symptoms.[16 17] Studies have proposed the use of ultrasound to detect tendon structural changes associated with the development of tendinopathy and potential injury to provide an opportunity for preventative programmes.[16 18 19] Importantly though, these

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Table 1 Demographics of athletes (N=695)

Variable	Frequency (%)	Ultrasound abnormality (%)		
		Patellar	Achilles	Plantar fascia
Sex				
Male	263 (38.9)	45.6	6.1	2.3
Female	414 (61.1)	31.0	8.2	3.1
Main sport				
Baseball	39 (5.6)	46.8	2.6	3.9
Basketball	139 (20.1)	53.2	8.0	5.0
Beach volleyball	35 (5.1)	52.9	4.3	0.0
Cross-country	58 (8.4)	21.6	22.4	7.8
Fencing	3 (0.4)	50.0	0.0	0.0
Field hockey	5 (0.7)	60.0	0.0	0.0
Football	2 (0.3)	0.0	25.0	0.0
Golf	3 (0.4)	0.0	0.0	0.0
Gymnastics	27 (3.9)	24.5	9.4	0.0
Lacrosse	65 (9.4)	35.4	7.7	0.0
Skiing	12 (1.7)	22.7	12.5	8.3
Soccer	27 (3.9)	39.2	0.0	1.9
Softball	39 (5.6)	24.7	0.0	1.3
Swimming/diving	68 (9.8)	5.9	2.2	0.0
Tennis	11 (1.6)	45.0	13.6	4.6
Track and field	108 (15.6)	33.2	6.1	2.8
Volleyball	50 (7.2)	59.8	12.0	2.0
Wrestling	2 (0.3)	25.0	0.0	0.0
Sport category				
Explosive	375 (54.1)	46.3	7.9	2.7
Running	166 (24.0)	29.1	11.8	4.5
Aquatic	68 (9.8)	5.9	2.2	0.0
Skill based	84 (12.1)	34.9	1.2	2.4
Age, years (mean (SD))	20.0 (1.6)			
Height in cm (mean (SD))	178.4 (12.1)			
Weight in kg (mean (SD))	74.2 (15.0)			
BMI in kg/m^2 (mean (SD))	23.1 (2.9)			
Years in sport (mean (SD))	9.4 (3.7)			

The percentage of structures (patellar tendons, Achilles tendons and plantar fasciae) with an ultrasound abnormality are also displayed.
BMI, body mass index.

Table 2 Numbers of tendon/fascia structures analysed in study (max 1390 tendons/fascia)

Variable	Frequency (%)
Number of left and right structures examined	
Patellar tendon*	1360 (100.0)
Achilles tendon†	1384 (100.0)
Plantar fascia‡	1389 (100.0)
Prior injury (yes)	
Patellar tendon (out of 1360)	50 (3.7)
Achilles tendon (out of 1384)	53 (3.8)
Plantar fascia (out of 1389)	39 (2.8)
Pain at time of scan (yes)	
Patellar tendon (out of 1360)	90 (6.6)
Achilles tendon (out of 1384)	58 (4.2)
Plantar fascia (out of 1389)	38 (2.7)

Tendons or fascia with previous surgery were excluded from the analysis.
*30 tendons were excluded due to history of surgery.
†6 tendons were excluded due to history of surgery.
‡1 fascia was excluded due to history of surgery.

studies have many shortcomings. Reliability of ultrasound evaluations remains unknown or low in numerous studies,[20–23] calling into question reproducibility of the findings. Other studies exhibited small or homogenous sample sizes,[20 21 24–26] suboptimal follow-up or injury monitoring[20 25 26] and short follow-up time frames.[27–29]

The present study aimed to determine if a preseason sonographic evaluation of patellar tendons, Achilles tendons and plantar fasciae, among a large cohort of collegiate athletes from a variety of sports, can identify associations with future time-loss injuries.

METHODS

Participants

Division I student-athletes were recruited each of 3 years (2021–2023) during their preparticipation physical examinations. All athletes of adult age were eligible to participate. Prior to study initiation, it was decided that athletes who sustained a partial or complete rupture to one of the tendons/fasciae had that particular structure excluded from analysis due to the potentially confounding underlying pathophysiology, as this study focused on non-acute (gradual) injuries, namely tendinopathy/fasciopathy.[32] Similarly, those with prior surgery or rupture to the affected structure also had that particular structure excluded from analysis. Subjects were allowed to participate for more than 1 year and were scanned at each of the 1, 2 or 3 years time points they participated. The presence of tendinopathy/fasciopathy symptoms was not an exclusion criterion.

Questionnaire

On completion of informed consent, each subject completed a brief questionnaire prior to ultrasound scans. Subjects self-reported age, sex, height, weight and prior years of competition within their sport. Subjects also received an anatomic diagram of the patellar tendons, Achilles tendons and plantar fasciae and were asked to self-report prior injury, current pain or prior surgery for any of these structures (including the lateralit(ies)). Subjects were allowed to ask any question to the medical/training staff concerning the questionnaire. All subjects were able to practise at the time of evaluation.

Preseason ultrasound scans

Experienced sports medicine physicians with at least 5 years of post-training experience performed or directly oversaw (sports medicine fellows with at least 3 years of training) the attainment of protocolised video captures of the bilateral Achilles tendons, patellar tendons and plantar fasciae of all subjects. No clinical findings were reported; these were simply video captures of the tendons/fasciae. The ultrasound videos were recorded with a 12–18 MHz linear transducer (Sonosite PX, Sonosite X-porte, Logiq E9 R7 and Konica Minolta Sonimage HS-1). Video captures contained no identifiable information, were approximately 10 s in duration and were performed in an identical manner for all participants as previously described.[33 34] Briefly, longitudinal and transverse views of patellar and Achilles tendons were obtained, scanning the length of each structure, to allow for orthogonal views. The plantar fasciae only had longitudinal views. Neovascularisation was obtained for all structures as well; thus, a total of 16 videos were collected for each subject (3 videos for each bilateral patellar and Achilles tendons and 2 videos for each plantar fascia). Investigators did not disclose the results of the ultrasound findings to the subject.

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Table 3 Contingency table of ultrasound abnormality and injury (N=695 athletes/1390 tendons)

Tendon	Ultrasound abnormality	Injury		aRR (95% CI)*	P value*
		Yes	No		
Patellar	Yes	45 (9.0)	453 (91.0)	8.9 (3.7, 21.4)	**<0.001**
	No	7 (0.8)	855 (99.2)		
Hypoechogenicity	Yes	43 (8.9)	440 (91.1)		
	No	9 (1.0)	866 (99.0)		
Thickening	Yes	36 (13.0)	240 (87.0)		
	No	16 (1.5)	1066 (98.5)		
Neovascularity	Yes	33 (17.5)	156 (82.5)		
	No	17 (1.5)	1142 (98.5)		
Achilles	Yes	17 (16.3)	87 (83.7)	18.8 (7.2, 48.8)	**<0.001**
	No	8 (0.6)	1272 (99.4)		
Hypoechogenicity	Yes	15 (18.8)	65 (81.2)		
	No	10 (0.8)	1292 (99.2)		
Thickening	Yes	11 (19.3)	46 (80.7)		
	No	14 (1.1)	1311 (98.9)		
Neovascularity	Yes	4 (20.0)	16 (80.0)		
	No	21 (1.5)	1335 (98.5)		
Plantar fascia	Yes	5 (12.8)	34 (87.2)	21.0 (6.4, 68.1)	**<0.001**
	No	4 (0.3)	1346 (99.7)		
Hypoechogenicity	Yes	5 (16.7)	25 (83.3)		
	No	4 (0.3)	1352 (99.7)		
Thickening	Yes	5 (14.7)	29 (85.3)		
	No	4 (0.3)	1348 (99.7)		
Neovascularity	Yes	3 (50.0)	3 (50.0)		
	No	6 (0.4)	1374 (99.6)		

Values are frequency (%).
Bolded values indicate statistical significance, $p < 0.05$.
*Generalised linear model for binomial distribution and log link with cluster-robust SEs, adjusting for prior injury and pain at time of ultrasound screening.
aRR, adjusted relative risk.

Blinded ultrasound video analysis

The primary author independently reviewed the videos of all structures for all subjects and was fully blinded to subject identity. Reviews were performed prior to the recording of any injury reports. The criteria used in the video review are listed in online supplemental table 1. Inter-rater reliability was tested on a subset of 80 of each structure with a second experienced sonographer after the end of the final year of data collection and showed near-perfect reliability (kappa values ranging from 0.947 to 1.000). The statistician was blinded to subject information.

Injury definition

Injuries were defined as (1) the development of pain in the Achilles tendon, patellar tendon or plantar fascia and (2) tenderness and load-related pain to the affected structure,[17 35] that resulted in (3) loss of time for practice or competition (one or more practices or competitions). Diagnoses were made by athletic training staff or board-certified sports medicine physicians. Staff were notified of the importance to report these injuries at the onset of the study for fidelity. All reported diagnoses that fit the aforementioned criteria were discussed with staff/physicians for accuracy; staff were also queried every 6 months about potentially missed diagnoses. As staff were already required to report these to the medical record system, no missed diagnoses were identified.

Deidentified data collected as part of the Pac-12 Health Analytics Program (HAP) were derived from clinical documentation in a HIPAA Compliant electronic medical record by sports medicine clinicians in the Pac-12 Conference. Data were deidentified using HIPAA Safe Harbor method for de-identification (45 CFR 164.514). Resulting project data included deidentified records only from athletes that provided authorisation for secondary research as part of the HAP.

Data analysis

Prior literature[17] suggested a risk ratio of approximately 4 for the patellar tendon and 7 for the Achilles tendon, suggesting a sample size of 141 (baseline risk=0.03; proportions of athletes with and without ultrasound abnormalities=0.25 and 0.75, respectively; α=0.05; power=0.80), or 177 subjects for the purpose of identifying how prior injury, as a predisposing factor, was involved in the development of sonographic tendon abnormalities (expected probabilities of developing sonographic tendon abnormalities with and without prior injury=25% and 5%, respectively; overall prevalence of prior injury=10%; α=0.05; power=0.80). However, as no prior literature was available for the plantar fascia, a goal of 600 athletes was decided for recruitment. Demographics of athletes were summarised with descriptive statistics. Data on ultrasound findings and injuries resulting in time loss were analysed per academic year (ie, person-years), separately for the patellar tendon, Achilles tendon and plantar fascia. It should be noted that some athletes were included over 2 or 3 years (eg, years 1, 2 and 3). Importantly, they received new ultrasound scans each year and their injuries were monitored for that subsequent year. If an athlete sustained multiple injuries on the same tendon/fascia in the same year, only the first injury was included in the analysis. The association of ultrasound finding to injury was analysed using a two-by-two contingency table analysis, with p values calculated from a generalised linear model (GLM) for binomial distribution and log link[36 37] with the cluster-robust variance estimator (unique athlete in each season as cluster variable).[38 39] The model included prior injury (categorical, dichotomous variable) and pain at the time of the scan (categorical, dichotomous variable) as covariates. These variables were chosen a priori[40–42] based on literature[4–10 43] and expertise from the clinician investigators and were considered the most relevant and practical to clinical practice for determining the risk for tendon injuries. An adjusted relative risk (aRR),[44 45] along with its 95% CI, was computed from the model to aid in interpretations. In addition, the Cox proportional hazards (PH) model[46] with the cluster-robust SEs (unique athlete in each season as cluster variable)[38 39] was used to analyse days until injury development by ultrasound abnormality. The origin and start times were the date when the athlete received ultrasound screening, whereas the end time was either the date when the athlete sustained an injury or end of the season. Similar to the GLM above, prior injury (categorical, dichotomous variable) and pain at the time of the scan (categorical, dichotomous variable) were included as covariates in the model. The assumption of PH in the Cox PH model was examined and ensured, as the interaction between time and each covariate was not statistically significant ($p>0.05$).[46 47] An HR and its 95% CI were calculated while including prior injury and pain at the time of the scan as covariates. Further, sensitivity (SE), specificity (Sp), positive predictive value (PPV) and negative predictive value (NPV) were calculated to evaluate ultrasound findings, prior injury and pain at the time of the scan as screening tools for injury development. CIs of these measures were computed based on exact/Clopper-Pearson binomial CIs.[48] All analyses meet CHAMP guidelines[31] and were performed by using Stata/MP V.18.0 (StataCorp), with

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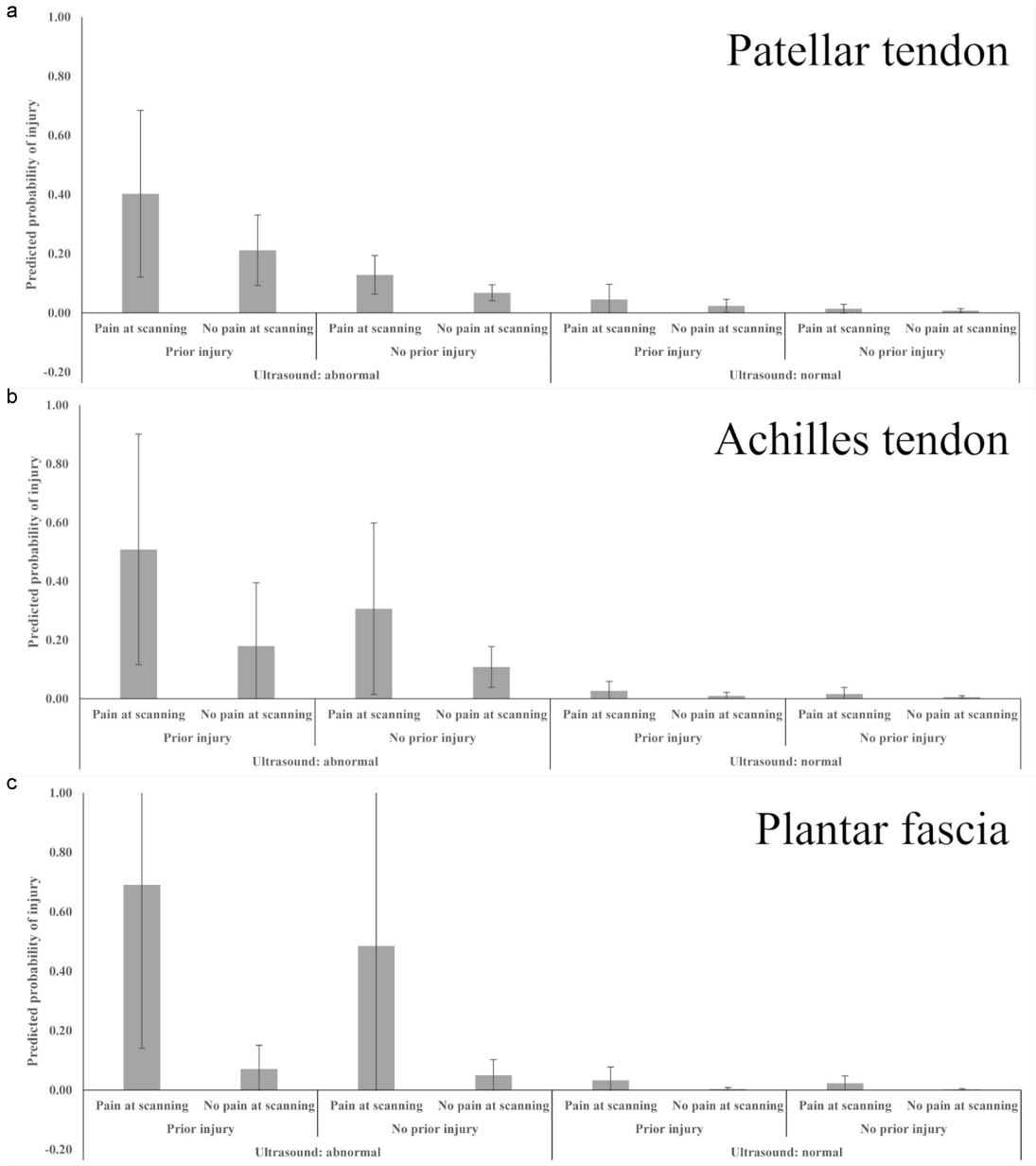

Figure 1 (A) Cox proportional hazards model with cluster-robust SEs on injury by ultrasound finding in patellar tendon adjusting for prior injury and pain at the time of ultrasound screening. (B) Cox proportional hazards model with cluster-robust SEs on injury by ultrasound finding in Achilles tendon adjusting for prior injury and pain at the time of ultrasound screening. (C) Cox proportional hazards model with cluster-robust SEs on injury by ultrasound finding in plantar fascia adjusting for prior injury and pain at the time of ultrasound screening.

an α level of 0.05 for statistical significance. Data generated by this research are available on reasonable request.

Equity, diversity and inclusion statement

Our study population included all eligible collegiate student athletes at three National Collegiate Athletic Association (NCAA) Division I institutions without any exclusion criteria based on gender, race/ethnicity/culture, socioeconomic level or marginalised groups. Our investigator and author team includes members of various genders, ethnicity, disciplines, experience levels and institutions. All data collection and analyses assumed similar equity to resources among all subjects.

Patient and public involvement

Subjects were recruited during their annual physical examinations for participation in the study and were informed of the objectives of the study at the time of enrolment. Ample time was allowed for study-related questions, with no additional burden or time required of participating subjects.

RESULTS

Demographics

Over 3 years, data from 695 athlete scans (538 distinct athletes; 403 athletes with 1 scan, 113 athletes with 2 scans and 22 athletes with 3 scans) including 1390 left-sided and right-sided structures were analysed in this study, out of an estimated 6300 student-athletes at the 3 institutions. Two athletes were excluded due to a rupture of an Achilles tendon during the study time-frame. Demographic data are shown in table 1, while the number of tendons/fasciae analysed is seen in table 2. Athletes from 18 different sports were included with 61.1% being female, age 20.0±1.6 and body mass index 23.1±2.9 kg/m^2.

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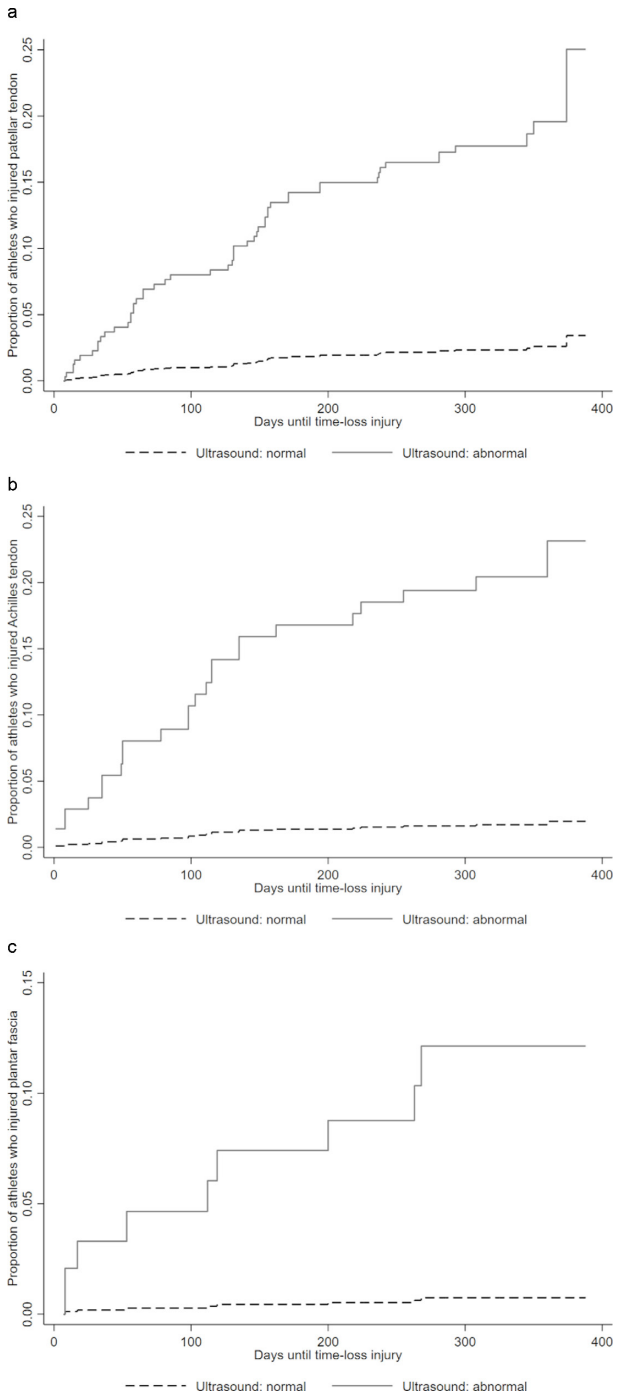

Figure 2 (A) Generalised linear model adjusting for ultrasound abnormality, prior injury and pain at the time of scan of the patellar tendon. (B) Generalised linear model adjusting for ultrasound abnormality, prior injury and pain at the time of scan of the Achilles tendon. (C) Generalised linear model adjusting for ultrasound abnormality, prior injury and pain at the time of scan of the plantar fascia.

Sonographic abnormalities and injury risk

Sonographic abnormalities were identified in 498 (36.6%) patellar tendons, 104 (7.5%) Achilles tendons and 39 (2.8%) plantar fasciae. Injury occurred at a frequency of 3.3% (44 injuries) in the patellar tendon, 1.6% (22 injuries) in the Achilles tendon and 0.7% (9 injuries) in the plantar fascia.

There was a significant association between the presence of a preseason ultrasound abnormality and the development of time-loss injury for all tendon/fascia structures in all athletes (p<0.001) (see table 3). Athletes with ultrasound abnormalities in the patellar tendon were about 9 times more likely to develop a time-loss injury (aRR=8.9; 95% CI 3.7, 21.4), 19 times more likely in the Achilles tendon (aRR=18.8; 95% CI 7.2, 48.8), 21 times more likely in the plantar fascia (aRR=21.0; 95% CI 6.4, 68.1). The hazards of developing a time-loss injury in the patellar tendon with an identified ultrasound abnormality were around 8 times higher than those without (HR=8.2; 95% CI 3.3, 20.5; p<0.001), 13 times higher in the Achilles tendon (HR=13.3; 95% CI 5.2, 34.1; p<0.001) and 17 times higher in the plantar fascia (HR=17.2; 95% CI 5.1, 58.2; p<0.001; figure 1). Ultrasound abnormalities in all three structures of asymptomatic athletes (those who self-reported no pain at the pre-season scan) also revealed a significant association with the development of a time-loss injury (patellar tendon aRR 10.7 (95% CI=4.1, 27.8), p<0.001; Achilles tendon aRR 19.2 (95% CI=6.6, 55.7), p<0.001; plantar fascia aRR 13.3 (95% CI 1.4, 104.8), p=0.025).

Analysis of injury predictors

Online supplemental table 2 and figure 2 summarise a GLM model evaluating three predictors of time-loss injury including ultrasound abnormality, prior injury and pain at the time of the scan. Ultrasound abnormalities of the patellar tendon were identified to be more predictive of time-loss injury than prior injury and pain at the time of the scan because of the size of RR, though all predictors were statistically significant (p<0.05). In the Achilles tendon, ultrasound abnormality was the only significant predictor of time-loss injury (p<0.001); whereas, both ultrasound abnormality and pain at the time of the scan were significant predictors of time-loss injury in the plantar fascia (p<0.001), despite the fairly wide 95% CIs of the coefficients.

Screening tool comparison

In terms of diagnostic abilities of the three screening tools for predicting injuries (table 4), SE was much higher for ultrasound screening than for prior injury or pain at the time of the scan for all tendon/fascia structures, which was statistically significant in most comparisons because of nonoverlapping 95% CIs. Sp was high for screening in the patella tendon and plantar fascia (>93%), but Sp for ultrasound screening in the patellar tendon was lower (65.4%). There was a wide variation of PPVs within each screening tool by tendon/fascia structure. NPV was close to 100% for all screening tools in any tendon.

DISCUSSION

Sonographic abnormalities most predictive of injury

This multi-institutional study of NCAA Division I athletes demonstrated that an identified preseason ultrasound abnormality in the patellar tendon, Achilles tendon or plantar fascia was more predictive of the development of a time-loss injury than other known risk factors. Prior studies have reported various risk factors associated with the development of lower extremity injury, including one of the most notable risk factors being prior injury.[4 5 8–10] Owens *et al* identified that military personnel with prior trauma or injury to the patellar tendon, Achilles tendon or plantar fascia had the highest adjusted OR for development of tendinopathy/fasciopathy within a 1-year time frame.[49] However, the findings of our current study clearly identified that the presence of a preseason ultrasound abnormality

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Table 4 Diagnostic summary measures of three screening tools on predicting injuries by tendon/fascia structure

		Diagnostic summary measure			
Screening tool	Tendon	Se (95% CI)	Sp (95% CI)	PPV (95% CI)	NPV (95% CI)
Ultrasound	Patellar	86.5 (74.2, 94.4)	65.4 (62.7 67.9)	9.0 (6.7, 11.9)	99.2 (98.3, 99.7)
	Achilles	68.0 (46.5, 85.1)	93.6 (92.2, 94.8)	16.3 (9.8, 24.9)	99.4 (98.8, 99.7)
	Plantar fascia	55.6 (21.2, 86.3)	97.5 (96.6, 98.3)	12.8 (4.3, 27.4)	99.7 (99.2, 99.9)
Prior injury	Patellar	21.2 (11.1, 34.7)	97.0 (95.9, 97.9)	22.0 (11.5, 36.0)	96.9 (95.8, 97.7)
	Achilles	24.0 (9.4, 45.1)	96.5 (95.4, 97.4)	11.3 (4.3, 23.0)	98.6 (97.8, 99.1)
	Plantar fascia	33.3 (7.5, 70.1)	97.4 (96.4, 98.2)	7.7 (1.6, 20.9)	99.6 (99.0, 99.8)
Pain at time of scan	Patellar	25.0 (14.0, 38.9)	94.1 (92.7, 95.3)	14.4 (7.9, 23.4)	96.9 (95.8, 97.8)
	Achilles	32.0 (14.9, 53.5)	96.3 (95.2, 97.3)	13.8 (6.1, 25.4)	98.7 (98.0, 99.3)
	Plantar fascia	44.4 (13.7, 78.8)	97.5 (96.6, 98.3)	10.5 (2.9, 24.8)	99.6 (99.1, 99.9)

NPV, negative predictive value; PPV, positive predictive value; Se, sensitivity; Sp, specificity.

was associated with a higher RR of the development of time-loss injury compared with prior injury. Importantly, differences in the military population (age, sex, level of activity, comorbidities, etc) may contribute to differences between studies. Youth basketball players with previous anterior knee pain had an 8.5 odds the risk of developing patellar tendinopathy over a season if they had prior knee pain,[43] while another study of subjects from a variety of demographic factors showed that a previous knee injury was associated with a 2.10 odds of having patellar tendinopathy.[50] For the Achilles tendon, prior tendinopathy was associated with a 6.3 increase in the odds of developing Achilles tendinopathy in recreational runners.[51]

Interestingly, ultrasound abnormalities were additionally more predictive of time-loss injury than the presence of pain at the time of scanning. This may be due to a high percentage of athletes having pain and simply playing through it, thus not having a time-loss injury. Or, self-report may not actually be accurate. Several conditions can have a similar pain location to tendinopathy/fasciopathy, including retrocalcaneal bursitis, flexor tendinopathy, patellofemoral pain syndrome, fat pad impingement, tarsal tunnel syndrome, entrapment of the first branch of the lateral plantar nerve and radiculopathy. For the purposes of this study, the authors purposefully chose to use self-report as it is the most realistic method of obtaining this information. A more accurate clinical assessment may improve the likelihood of painful tendons/fasciae progressing to future time-loss injury.

Prior studies have prospectively evaluated the likelihood of developing injury based on ultrasound findings in smaller cohorts and specific athlete/population groups.[16 17 25 26 29 34 52] Overall, evidence from these studies and systematic reviews has demonstrated that ultrasound abnormalities are associated with an increased risk of injury development. This current study is the largest of its kind, prospectively evaluating a cohort of NCAA Division I athletes at multiple institutions over 3 years of data collection. The results are consistent and support previous findings identifying a significant and comparatively higher RR for the development of injury among the three structures (eg, patellar tendon, Achilles tendon and plantar fascia). The higher RR may relate to our study population as this study included only NCAA Division I athletes, while other studies included individuals of various ages and backgrounds.[16 17] However, the majority of prior studies also examined asymptomatic tendons, while this study examined all tendons, adjusting for pain at the time of the scan.

Sonographic abnormalities were identified most commonly in the patellar tendon (36.6%) with associated time-loss injury occurring at a frequency of 3.3%. These findings are consistent with previous studies as the patellar tendon is known to show a higher percentage of sonographic abnormalities on preseason screening evaluations compared with Achilles tendon.[53] The plantar fascia had the lowest percentage of preseason ultrasound abnormalities in this study cohort (2.8%) with the frequency of injury occurring at 0.7%. Limited data are present in the literature regarding the relationship of identified ultrasound abnormalities and the development of future injury in the plantar fascia.[54 55] However, this study again supports that the presence of an ultrasound abnormality in any of the three structures evaluated is associated with a significant risk of the development of injury. The timing of these injuries appeared at different times throughout the year, with tendon injuries appearing to level off around 6 months; this may be related to differences in sport-specific competitive seasons or tendon remodelling occurring over this time frame.[56]

Screening with ultrasound

As a screening tool, ultrasound showed extremely high NPV (>99%) indicating that athletes with normal ultrasound findings were unlikely to develop an injury in the tendon/fascia structure. NPV was high among all screening tools given the relatively low prevalence of injury, but ultrasound scanning was slightly superior to that of prior injury and pain at the time of scanning among all three structures. Ultrasound screening also showed superior SE to the other screening tools in all three structures, which is an important characteristic of a screening tool in general.[57] PPVs were relatively low among all three screening tools in all three structures, indicating that most athletes who were positive on any screening tool did not develop an injury, as previously evidenced.[16]

Clinical implications

This study supports the notion that sonographic abnormalities in tendon/fascia structures are a significant risk factor for the development of injury and lays the groundwork for prospective preventative studies involving at-risk athletes. Interventional trials are needed and should follow this study to help prevent future injury; only one prior study has evaluated this topic and unfortunately failed to prevent injuries.[19] The authors suggest a thorough evaluation of at-risk athletes, examining neuromuscular control, strength, balance and range of motion of the lower extremities. Identification of biomechanical deficits could then be targeted with preventive rehabilitation, focusing on the deficits rather than just the abnormal structure(s). Additionally, current study findings should provide heightened awareness for

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clinicians of the risk for the potential for future injury in athletes with sonographic abnormalities. Clinicians may consider identifying and addressing concomitant workload, biomechanical or neuromuscular factors that may have contributed to such structural changes.

Limitations

While this study evaluated athletes from 18 different sports, they were not equally represented (eg, 20.1% men's basketball, while 3.9% gymnastics). Second, despite a large sample size including various sport disciplines, further studies are required to identify whether it is generalisable to other populations. Different age groups and sport types may have differing results. Third, ultrasound findings in this study were recorded by the presence or absence of an abnormality, rather than specifying by morphological type. Future studies could be considered to evaluate which specific abnormal sonographic findings are more likely to be associated with the development of injury. Fourth, recent studies have demonstrated that an anatomic diagram may be useful for Achilles-related pain,[58] but not patellar tendon[59]; thus, self-reported pain may not be as accurate as clinical assessment. Fifth, although the reliability was high among study team members, lack of quantifiable values for thickness and hypoechogenicity may affect utilisation of this method elsewhere. Sixth, clinical diagnoses can be challenging without a gold-standard method; athletic trainers and physicians may not have accurately diagnosed all injuries. Seventh, the duration of time-loss was not adequately captured; thus, relatively minor amounts of time-loss may have been included equally to severe time-loss. Eighth, relatively few injury cases in this study (3.3% or 44 injuries in the patellar tendon; 1.6% or 22 injuries in the Achilles tendon; 0.7% or 9 injuries in the plantar fascia) resulted in wider CIs for risk measures, which could have introduced sparse data bias.[60 61] Ninth, since this was an observational study, it was possible that unmeasured/unobserved confounding factors might have affected the study results.[62] Lastly, injuries to the tendon/fascia structures were described as either 'yes' or 'no'; additional specifics on type of injury and duration of time-loss could be helpful to understand injury and recovery patterns associated with ultrasound findings. Importantly, utilisation of this technique should be interpreted with caution in other populations (such as adolescents and older athletes).

Strengths

Despite the identified limitations, this study has multiple strengths that should be highlighted. This study is the largest of its kind and included athletes from three different division I institutions. Prior studies have focused on individual groups of athletes (eg, basketball, soccer, endurance athletes),[16 17] while this study evaluated athletes from a broad number of sports collectively allowing for more robust generalisations of collegiate athletes. Athletes were monitored closely with the help of athletic training staff, coaches and team physicians over the duration of this study. Data collection was excellent for the 695 athletes included in this study with only 37 missing data points, which accounts for less than 1% of the overall data.

CONCLUSION

The identification of sonographic abnormalities in the patellar tendon, Achilles tendon or plantar fascia on preseason evaluation was identified to have a higher association with the risk of developing time-loss injury than self-reported prior injury or even the presence of self-reported pain at the time of the scan. Preseason ultrasound screening should be considered in collegiate athletes to evaluate for the risk of developing injury during their sport season. Importantly, further research should be focused on specifying sonographic abnormalities with higher risk and identifying injury prevention interventions.

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Patient and public involvement Patients and/or the public were involved in the design, or conduct, or reporting, or dissemination plans of this research. Refer to the Methods section for further details.

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